

Power Automate Curriculum



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WEEK 1: FOUNDATIONS & BASICS

Goal: Understand the interface, triggers, and building simple "A-to-B" flows.

- **Topic 1: SharePoint as a Data Source.**

Lists vs. Libraries: When to use each (Structured data vs. Document management).

Column Types: Understanding Choice, Date, Person, and Lookup columns (and why "Multiple lines of text" can be tricky in flows).

- **Topic 2: The Big Picture.**

Introduction to the Power Platform ecosystem. Explain the difference between Automated, Instant, and Scheduled flows.

- **Topic 3: Copilot & Templates.**

Teaching how to use Microsoft Copilot to describe a flow in natural language.

- **Topic 4: SharePoint Integration (The Basics).**

Triggers like "When an item is created" and actions like "Send an email (V2)."

- **Practice:**

Build a flow that notifies a Teams channel whenever a new document is uploaded to a specific SharePoint folder.

WEEK 2: LOGIC, LOOPS, AND APPROVALS

Goal: Adding "intelligence" to flows using conditions and standard business processes.

- **Topic 1: Conditional Logic.**

Using the Condition (If/Yes/No) and Switch actions to handle different data values.

- **Topic 2: The Approvals Connector.**

Mastering "Start and wait for an approval." Discussing the difference between Basic and Custom approvals.

- **Topic 3: Working with Lists (Loops).**

Introduction to the Apply to Each loop and how to update multiple SharePoint items at once.

- **Practice:**

Create a "Leave Request" system: User submits a SharePoint list item -> Manager gets a Teams Approval -> List status updates to "Approved" or "Rejected."

WEEK 3: DATA MANIPULATION & EXPRESSIONS

Goal: Move beyond the "Dynamic Content" picker and start using formulas.

- **Topic 1: Introduction to WDL (Workflow Definition Language).**

Using basic expressions like utcNow(), formatDateTime(), and variables.

- **Topic 2: Data Operations.**

Using Compose, Select, and Filter Array to clean data without using heavy loops.

- **Topic 3: Input/Output Mastery.**

Handling JSON data and using the Parse JSON action to make external data usable.

- **Practice:**

Build a "Weekly Digest" flow that finds all SharePoint items created in the last 7 days, filters for "High Priority" only, and emails a formatted HTML table to a supervisor.

WEEK 4: PROFESSIONALISM, ERROR HANDLING, AND AI

Goal: Building flows that don't break and following organization best practices.

- **Topic 1: Error Handling (Try-Catch).**

Using Configure Run After to send a notification if a flow fails.

- **Topic 2: Advanced SharePoint & AI.**

Using AI Builder to extract data from an invoice (PDF) and save it to SharePoint.

- **Topic 3: Solutions & Governance.**

Why we build inside Solutions and naming conventions (e.g., [Dept] - [Trigger] - [Action]).

- **Practice:**

A "Travel Expense Claim" flow that uses AI to read a receipt, asks for approval if the amount is >\$100, and logs the final result in a SharePoint Archive.

WEEK 5: THE PROFESSIONAL HANDOFF (ALM)

Goal: Learn how to move automations between environments (Dev -> Test -> Prod) and manage dependencies.

- **Topic 1: The Container Concept (Solutions).**

Why we don't use "My Flows."

- Creating a Solution and adding existing flows, SharePoint connection references, and environment variables.

- **Topic 2: Exporting & Packaging.**

The difference between Managed (read only for Prod) and Unmanaged (editable for Dev) solutions.

- Checking for "Missing Dependencies" before exporting.

- **Topic 3: Importing & Connection References.**

The Import process: Mapping connections to the new environment.

- Using Environment Variables so SharePoint Site URLs change automatically when moving from Dev to Prod.

- **Practice:**

Export the "Travel Expense" flow from the Developer environment and successfully import it into a "Test" environment, ensuring all SharePoint links update correctly without manual editing.

